ANSWER KEY

Lab 6 HTTP plain text credentials

Capture the traffic in Wireshark (Windows 11 PC, sign in with the password networkwalks) and follow the solution steps below.

Task 1. Capture the traffic — 10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Answer: 255

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 255.

Task 2. Which protocol — 10 marks

A staff member signed in to the old staff portal. Which protocol carried the sign-in?

Answer: HTTP

Solution: Filter

```
http
```

. The sign-in packets say POST /login HTTP/1.1. The protocol is HTTP, which is not encrypted.

Task 3. The method — 10 marks

A web form sends the typed values to the server. Which HTTP method did the sign-in use?

Answer: POST

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The method is POST.

Task 4. The login page — 10 marks

Which page path did the form send the sign-in to?

Answer: /login

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The path is /login.

Task 5. The PC that signed in — 10 marks

Which PC sent the sign-in?

Answer: 192.168.10.45

Solution: Filter

```
http.request.method == "POST"
```

. The Source column shows 192.168.10.45.

Task 6. How many attempts

10 marks

How many times did the user send the sign-in form?

Answer: 2

Solution: Filter

```
http.request.method == "POST"
```

. Two packets stay in the list. The user tried 2 times.

Task 7. The username

10 marks

Click one POST packet. In the packet details, open HTML Form URL Encoded. What is the username?

Answer: farhan.k

Solution: Click a POST packet and open HTML Form URL Encoded. The line

```
username = farhan.k
```

shows the user.

Task 8. The password

10 marks

The second sign-in worked (the page said “Welcome back”). What password did the user type that time?

Answer: Mango#2026

Solution: Click the second POST packet and open HTML Form URL Encoded. It shows

```
password = Mango#2026
```

. The first try (Mango2025) failed.

Task 9. Packet details line

10 marks

Which line in the packet details shows the typed username and password in an easy list?

Answer: HTML Form URL Encoded

Solution: At the bottom of the packet details you find HTML Form URL Encoded. It lists the username and password.

Task 10. The fix

10 marks

How can the company stop people from reading passwords like this?

Answer: Use HTTPS

Solution: HTTPS encrypts the traffic, so a sniffer sees only unreadable data. The answer is Use HTTPS.

SUMMARY

Protocol:	HTTP, POST /login
User and PC:	farhan.k from 192.168.10.45
Logins:	2 (1 failed, 1 worked)
Password:	Mango#2026
Fix:	Use HTTPS

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro